

Machine Learning Basics

What is Machine Learning (ML)?

Machine Learning is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

Traditional Programming

Data + Rules → Output

Machine Learning

Data + Output → Model (Rules Learned Automatically)

Example:

- Traditional: You write rules to detect spam emails.
 - ML: You provide thousands of spam and non-spam emails, and the model learns the rules itself.
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Types of Machine Learning

1. Supervised Learning

The model learns using **labeled data** (input + correct output).

A. Regression

Purpose: Predict a **continuous numerical value**.

Examples:

- House price prediction
- Temperature prediction
- Stock price prediction

- Salary prediction

Algorithms

- Linear Regression
- Ridge Regression
- Lasso Regression
- Decision Tree Regressor
- Random Forest Regressor

B. Classification

Purpose: Predict a **category or class**.

Examples:

- Spam vs Not Spam
- Cancer vs No Cancer
- Pass vs Fail
- Dog vs Cat

Algorithms

- Logistic Regression
 - Decision Tree
 - Random Forest
 - Naive Bayes
 - SVM
 - Neural Networks
-

2. Unsupervised Learning

The model learns from **unlabeled data** and finds hidden patterns.

A. Clustering

Groups similar data points together.

Examples:

- Customer segmentation
- Grouping similar news articles
- Friend suggestions on social media

Algorithms:

- K-Means
- DBSCAN
- Hierarchical Clustering

C. Dimensionality Reduction

Reduces the number of input features while preserving most of the important information.

Example:

- Faster training
- Less memory usage
- Easier visualization
- Reduces overfitting

Algorithms:

- PCA (Principal Component Analysis)
- t-SNE
- UMAP

3. Reinforcement Learning

An agent learns by interacting with an environment and receiving rewards or penalties.

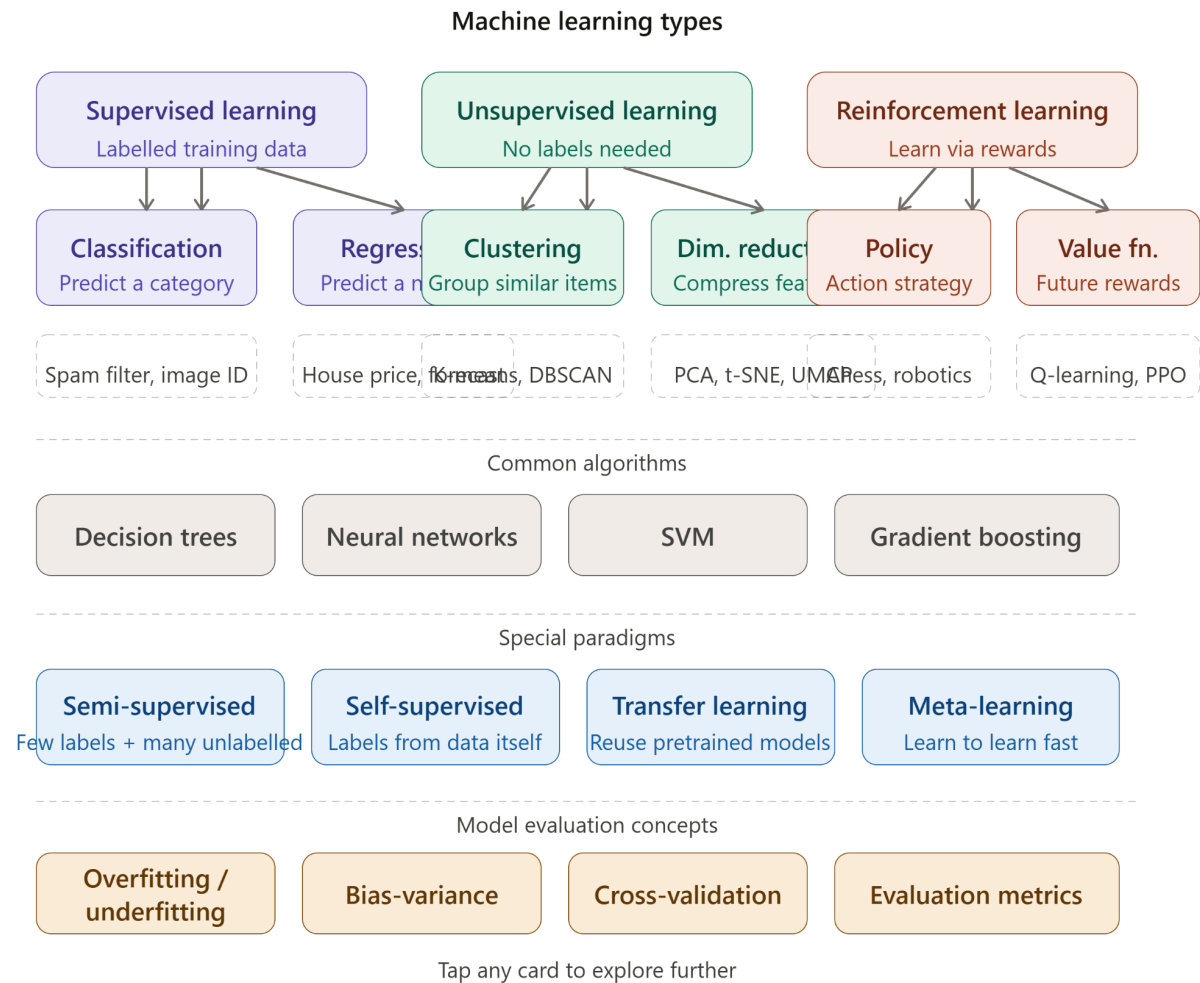
Examples

- Self-driving cars
- Chess engines
- Game AI

Concepts

- Agent
- Environment
- Reward

- Action



Why is Machine Learning Used?

ML is used when writing rules manually becomes difficult.

Uses of ML

Prediction

- Stock prices
- Weather forecasting
- Sales forecasting

Classification

- Spam detection
- Disease diagnosis
- Fraud detection

Recommendation Systems

- Netflix recommendations
- YouTube suggestions
- Amazon products

Computer Vision

- Face recognition
- Object detection
- Medical image analysis

Natural Language Processing (NLP)

- ChatGPT
 - Translation
 - Sentiment analysis
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Machine Learning Workflow

1. Collect Data



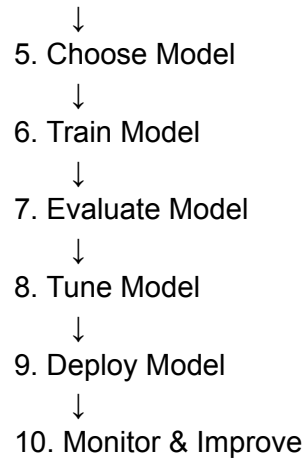
2. Clean Data



3. Feature Engineering



4. Split Dataset



Step 1: Data Collection

Gather data from:

- Databases
 - APIs
 - Surveys
 - Sensors
 - Websites
-

Step 2: Data Cleaning

Remove:

- Missing values
 - Duplicate data
 - Incorrect data
-

Step 3: Feature Engineering

Convert raw data into useful features.

Example

Raw Data:

Date = 11-07-2026

Features:

Day = Saturday

Month = July

Weekend = Yes

Step 4: Train-Test Split

Common split:

- 80% Training
- 20% Testing

Training data teaches the model.

Testing data checks performance.

Step 5: Model Selection

Choose an algorithm:

- Linear Regression
 - Decision Tree
 - Random Forest
 - XGBoost
 - Neural Networks
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Step 6: Training

The model learns patterns from training data.

Step 7: Evaluation

Regression Metrics

- MAE
- MSE
- RMSE
- R^2 Score

Classification Metrics

- Accuracy
 - Precision
 - Recall
 - F1 Score
-

Step 8: Hyperparameter Tuning

Improve performance by adjusting settings.

Examples:

- Number of trees
 - Learning rate
 - Maximum depth
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Step 9: Deployment

Make the model available through:

- Web applications
 - Mobile apps
 - APIs
-

Step 10: Monitoring

Check:

- Accuracy
- Performance

- Data drift

Retrain if needed.

Machine Learning vs Data Science vs Data Analyst

Aspect	Data Analyst	Data Science	Machine Learning
Goal	Understand past data	Extract insights & predict	Build intelligent systems
Focus	Reports & dashboards	Statistics + Programming	Algorithms & Models
Tools	Excel, SQL, Power BI	Python, SQL, Statistics	Python, TensorFlow, PyTorch
Output	Business insights	Predictions & insights	Trained models
Math Level	Low-Medium	Medium-High	High
Coding Level	Low-Medium	Medium-High	High

Data Analyst

Answers:

"What happened?"

Example:

- Monthly sales report
- Revenue dashboard

Data Scientist

Answers:

"Why did it happen and what will happen next?"

Example:

- Predict future sales
 - Customer churn analysis
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Machine Learning Engineer

Answers:

"How can we automate predictions at scale?"

Example:

- Recommendation systems
 - Fraud detection models
 - Chatbots
-

Statistics & Mathematics for Machine Learning

These topics form the mathematical foundation of Machine Learning. Understanding them will help you grasp algorithms like Linear Regression, Decision Trees, Neural Networks, and more.

1. Mean (Average)

The **mean** is the average of all observations.

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Where:

- (x) = each value
- (n) = number of values

Example

Data:

2, 4, 6, 8, 10

Mean:

$$\frac{2 + 4 + 6 + 8 + 10}{5} = 6$$

Answer = 6

Applications in ML

- Data normalization
- Missing value replacement
- Calculating variance & standard deviation
- Finding the center of data

2. Median

The **median** is the middle value after sorting the data.

Formula

Formula

If n is odd

$$Median = \left(\frac{n+1}{2} \right)^{th} \text{ value}$$

If n is even

$$Median = \frac{\frac{n}{2}^{th} + \left(\frac{n}{2} + 1 \right)^{th}}{2}$$

Example (Odd)

1 3 5 7 9

Median = 5

Example (Even)

2 4 6 8

Applications

- Handles outliers better than mean
 - Income analysis
 - House price prediction
 - Robust data preprocessing
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3. Mode

The **mode** is the value that appears most frequently.

Example

1 2 2 3 4 4 4 5

Mode = 4

Applications

- Categorical data
- Customer preference analysis
- Recommendation systems

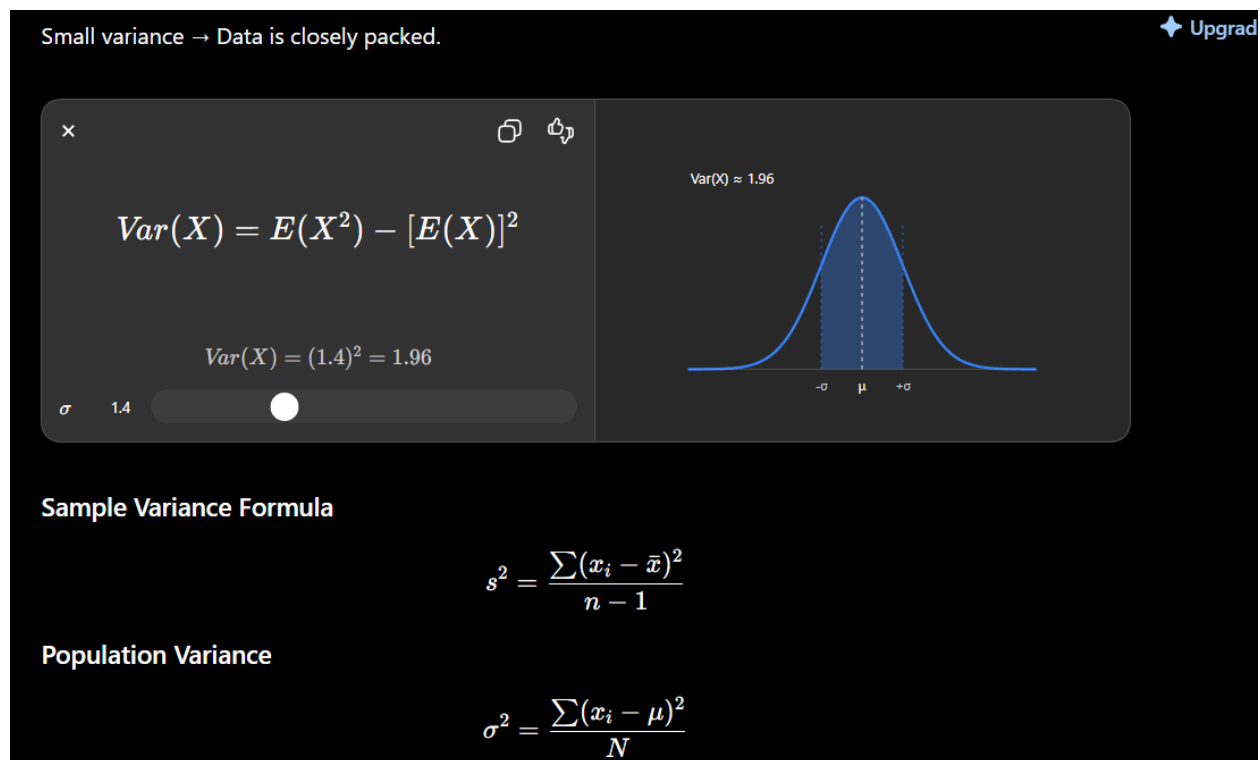
- Missing categorical values

4. Variance

Variance measures how far data points are spread from the mean.

Large variance → Data is widely spread.

Small variance → Data is closely packed.



Applications

- Feature selection
- Detecting data spread
- PCA
- Data preprocessing

5. Standard Deviation (SD)

Standard deviation is the square root of variance.

Formula

Formula

$$\sigma = \sqrt{\sigma^2}$$

Example

Variance = 25

$$\sqrt{25} = 5$$

Applications

- Standardization (Z-score)
 - Outlier detection
 - Risk analysis
 - Model evaluation
-

6. Range

Range is the difference between the maximum and minimum values.

Formula

Formula

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

Applications

- Understand data spread
 - Initial exploratory data analysis (EDA)
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7. Quartiles

Quartiles divide sorted data into **4 equal parts**.

Q1	Q2	Q3
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25%	50%	75%

Q1 (First Quartile)

25% of data lies below Q1.

Q2 (Second Quartile)

Median

Q3 (Third Quartile)

75% of data lies below Q3.

Interquartile Range (IQR)

$$IQR = Q3 - Q1$$

Applications:

- Outlier detection
 - Data preprocessing
 - Box plots
-

Summary Table

Measure	Meaning	Formula
Mean	Average	$\Sigma x / n$
Median	Middle value	Depends on n
Mode	Most frequent value	Highest frequency
Variance	Spread of data	Average squared deviation
Standard Deviation	Spread in original units	$\sqrt{\text{Variance}}$
Range	Overall spread	Max – Min
Quartiles	Divide data into four parts	Q1, Q2, Q3
IQR	Middle 50% spread	Q3 – Q1

8. Vectors

A **vector** is an ordered list of numbers representing magnitude and direction (or features).

Example:

Example:

$$A = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

Applications:

- Representing features
 - Word embeddings
 - Images
 - Neural networks
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Vector Addition

Vectors must have the same dimensions.

Vectors must have the same dimensions.

$$A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, B = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$A + B = \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

9. Matrices

A matrix is a rectangular arrangement of numbers.

Example

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Applications

- Image processing
 - Deep Learning
 - Linear Regression
 - PCA
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Matrix Addition

Same size matrices.

Same size matrices.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

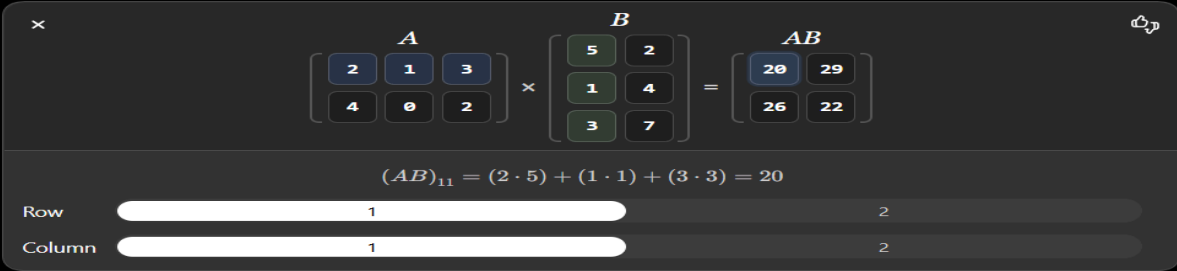
$$A + B = \begin{bmatrix} 6 & 8 \\ 10 & 12 \end{bmatrix}$$

Matrix Multiplication

Condition:

Columns of A = Rows of B

Applications:



Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$
$$AB = \begin{bmatrix} 19 & 22 \\ 43 & 50 \end{bmatrix}$$

- Neural networks
- Image transformations
- Computer vision
- Data compression

10. Probability

Probability measures how likely an event is to occur.

Formula

Formula

$$P(E) = \frac{\text{Favourable Outcomes}}{\text{Total Outcomes}}$$

Example:

Rolling a die

Probability of getting 4

$$P(4) = \frac{1}{6}$$

Independent Events

One event does **not** affect the other.

Example:

- Tossing a coin
- Rolling a die

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cap B) = P(A)P(B) = (0.60)(0.45) = 0.27$$

$P(A)$ 0.60

$P(B)$ 0.45

$P(A) = 0.60$
 $P(B) = 0.45$

Example:

Coin:

$$P(H) = \frac{1}{2}$$

Die:

$$P(6) = \frac{1}{6}$$

Both occurring:

$$P(H \cap 6) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$$

Applications:

- Random sampling
- Simulations
- Naive Bayes (under independence assumptions)

Dependent Events

One event **affects** the probability of another.

Example:

Drawing two cards from a deck **without replacement**.

Example:

Drawing two cards from a deck without replacement.

First card:

$$P(A) = \frac{4}{52}$$

After one Ace is drawn:

$$P(B|A) = \frac{3}{51}$$

Conditional probability:

$$P(A \cap B) = P(A) \times P(B|A)$$

Applications:

- Medical diagnosis
 - Fraud detection
 - Customer behavior analysis
 - Sequential predictions
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